

Cubic Talbot Revivals and Fixed-Mode Geometry for the Periodic Airy Flow

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Abstract

We give an all-time operator atlas for the periodic Airy equation. Its Fourier flow is a strongly continuous unitary group with least full-space period 2π . At every reduced rational time $2\pi p/q$, the propagator is an exact cubic discrete-Fourier sum of q spatial translations and has global order q .

1 Frozen flow and the global clock

On $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ fix

$$u_t + u_{xxx} = 0, \quad e_n(x) = e^{inx}. \quad (1)$$

Writing $D = -i\partial_x$, Fourier evolution gives

$$U(t)e_n = e^{in^3 t} e_n = e^{itD^3} e_n. \quad (2)$$

The real Fourier multiplier n^3 makes D^3 self-adjoint on its Sobolev domain. Parseval and dominated convergence therefore prove that $U(t)$ is a strongly continuous unitary group on $L^2(\mathbb{T})$.

If $U(t) = I$, the mode $n = 1$ forces $t \in 2\pi\mathbb{Z}$; conversely every integer cube has integral phase at such times. Thus 2π is the least positive return of the *whole* phase space. This is distinct from the often shorter return of an individual state.

2 Complete rational-revival theorem

Theorem 1. *Let $(p, q) = 1$ and $q \geq 1$. At $t = 2\pi p/q$,*

$$U(t)f = \sum_{r=0}^{q-1} A_{p,q}(r) \tau_{2\pi r/q} f, \quad A_{p,q}(r) = \frac{1}{q} \sum_{s=0}^{q-1} e^{2\pi i(p s^3 - sr)/q}, \quad (3)$$

where $\tau_a f(x) = f(x + a)$. Moreover

1. $\sum_r |A_{p,q}(r)|^2 = 1$ and the strobe has exact order q ;
2. its fixed space is

$$\overline{\text{span}}\{e_n : q \mid n^3\} = \overline{\text{span}}\{e_n : L(q) \mid n\}, \quad L(q) = \prod_{\ell^a \parallel q} \ell^{\lceil a/3 \rceil};$$

3. if $t/(2\pi)$ is irrational, the fixed space of $U(t)$ consists exactly of constants.

Proof. The identity

$$(n + q)^3 - n^3 = q(3n^2 + 3nq + q^2)$$

makes the multiplier in (2) q -periodic in n . Finite Fourier inversion yields (3); its unit-modulus input vector gives Parseval. The q th power is the identity, while the mode-one phase forces any order to be divisible by q . Finally, $q \mid n^3$ is equivalent prime by prime to $v_\ell(n) \geq \lceil v_\ell(q)/3 \rceil$. At irrational time, fixing a nonzero mode would make $n^3 t/(2\pi)$ an integer, a contradiction.

The baseline stops with the all-rational revival theorem. Fixed-support state periods, executable receipts, and operator boundaries are added only in later revisions.

3 Conclusion and limitations

One theorem now controls the complete periodic Airy group, every rational revival, every sampled fixed subspace, and every finite-support period. It does not classify nonperiodic third-order boundary conditions, evaluate all cubic Gauss sums in closed form, or extend to nonlinear KdV.

References

- [1] B. Pelloni and D. A. Smith, “Revivals, or the Talbot effect, for the Airy equation,” *Studies in Applied Mathematics* (2024), doi:10.1111/sapm.12699.
- [2] G. Farmakis et al., “Beyond periodic revivals for linear dispersive PDEs,” *Proceedings of the Royal Society A* 477 (2021), doi:10.1098/rspa.2021.0241.